AP Chemistry

Summer Work

Naming & Writing the Formulas of Chemical Compounds

Naming Simple Compounds

Molecule or **molecular compound** is an assembly of two or more non-metal atoms tightly bonded together.

A **diatomic molecule** is a molecule made from two atoms of the same element. The following seven elements form diatomic molecules in their natural state:

H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2

An **allotrope** is one of two or more distinct forms of an element, such as: graphite and diamond (for carbon) and dioxygen (O_2 - usually referred to simply as oxygen) and ozone (O_3).

Molecular compounds (also known as **covalent compounds)** are composed of nonmetal elements that bond together into larger particles using covalent bonds (bonds created by the sharing of their valence electrons.

> **Molecular formulas** show the exact number of atoms of each element in the molecule.
>
> **Empirical formulas** are the simplest whole-number ratio of the atoms in a molecular compound or an ionic compound...

Ionic compounds are composed of ions and usually contain both metals and non-metals. The ions in an ionic compound form when the metal atoms give one or more electrons to the nonmetal atoms. Consequently, the metal ion is positively charged (called an cation) and the nonmetal ions are negatively charged (called the anions). Ionic compounds must be electrically neutral, so the sum of the charges of the anions and cations must equal zero. Because the formulas of ionic compounds must be empirical formulas, make sure the subscripts are reduced to their simplest ratio.

Watch the following instructional video:

Ionic vs. Molecular Compounds

https://www.youtube.com/watch?v=PKA4CZwbZWU

Naming & Writing Formulas of Molecular Compounds

Watch the following instructional video:

https://www.youtube.com/watch?v=DejkvR4pvRw

Identifying a Molecular Compound (aka Covalent Compound)

A molecular compound **contains ALL non-metal elements** (located on the right side of the staircase on the periodic table).

General Naming Rule:

Prefix (except mono) + nonmetal name + prefix + nonmetal name + ide

Prefix Definitions

Prefix	Meaning	Prefix	Meaning
Mono	1	Hexa	6
Di	2	Hepta	7
Tri	3	Octa	8
Tetra	4	Nona	9
Penta	5	Deca	10

Examples of molecular compounds

Formula	Name
CO	carbon monoxide
CO_2	carbon dioxide
NO_2	nitrogen dioxide
N_2O_4	dinitrogen tetraoxide

Naming & Writing Formulas for Molecular Compounds

Practice Problems

Instructions: Write the formula from the names of the following molecular compounds and vice versa.

1. **di**silicon **tri**oxide _____
2. nitrogen **di**oxide _____
3. carbon **tetra**chloride _____
4. **tri**sulfur **mono**xide _____
5. phosphorus **tri**sulfide _____
6. boron tribromide _____
7. carbon monoxide _____
8. dinitrogen pentoxide _____
9. carbon dioxide _____
10. phosphorus hexafluoride _____
11. _____ $SeCl_2$
12. _____ NO
13. _____ P_2S
14. _____ Si_3I_8
15. _____ S_2O
16. _____ PBr_5
17. _____ N_2Se_3
18. _____ As_2O_4
19. _____ PH_3
20. _____ OCl_2

Naming & Writing Formulas for Molecular Compounds

ANSWERS

Practice Problems

Instructions: Check your answers.

1. **di**silicon **tri**oxide — Si_2O_3
2. nitrogen **di**oxide — NO_2
3. carbon **tetra**chloride — CCl_4
4. **tri**sulfur **mono**xide — S_3O
5. phosphorus **tri**sulfide — PS_3
6. boron tribromide — BBr_3
7. carbon monoxide — CO
8. dinitrogen pentoxide — N_2O_5
9. carbon dioxide — CO_2
10. phosphorus hexafluoride — PF_6
11. selenium dichloride — $SeCl_2$
12. nitrogen monoxide — NO
13. diphosphorus monosulfide — P_2S
14. trisilicon octaiodide — Si_3I_8
15. disulfur monoxide — S_2O
16. phosphorus pentabromide — PBr_5
17. dinitrogen triselenide — N_2Se_3
18. diarsenic tetraoxide — As_2O_4
19. phosphorus trihydride — PH_3
20. oxygen dichloride — OCl_2

Naming & Writing Formulas of Ionic Compounds

Identifying an Ionic Compound

An ionic compound will ***start with a metal element*** or ammonium (NH_4) and end with a non-metal or polyatomic ion (more about that later).

There are **three main types of ionic compound** to be considered:

- Basic (**binary**) ionic compounds (contain 1 metal & 1 nonmetal element)
- Ionic compounds containing a **polyatomic ion**
- Ionic compounds containing a **transition metal**

Examples of ionic compounds

Formula	Name
$BaCl_2$	barium chloride
$(NH_4)_2SO_4$	ammonium sulfate
Cu_2S	copper I sulfide
$Mn(NO_3)_4$	manganese IV nitrate

We will learn the rules for naming and writing formulas for each type, one at a time, and then put it all together.

Naming & Writing Formulas of Binary* Ionic Compounds

*a binary ionic compound is the simplest kind of ionic compound. It has only two elements in it (one metal and one nonmetal).

Naming Binary Ionic Compounds

General Rule:

1. Write the name of the metal ion (same as the element's name)
2. Write the name of the nonmetal ion (element's name, but change the ending to –ide)
   Example: $MgCl_2$ Magnesium Chlor~~ine~~ → Magnesium Chloride

Writing Formulas for Binary Ionic Compounds

Watch the following instructional videos:

Introduction: https://www.youtube.com/watch?v=URc75hoKGLY&t=19s

Practice: https://www.youtube.com/watch?v=X_LVANMpJ0c

Practice Problems

Instructions: Write the formula from the names of the following binary ionic compounds and vice versa.

1. _____ Na_2S
2. _____ SrO
3. _____ Be_3N_2
4. _____ CaF_2
5. _____ KCl
6. Cesium Phosphide _____
7. Barium Bromide _____
8. Lithium Oxide _____
9. Aluminum Selenide _____
10. Rubidium Iodide _____

Naming & Writing Formulas of Binary Ionic Compounds

Answers

Practice Problems

Instructions: Check your answers.

1. Sodium sulfide — Na_2S
2. Strontium oxide — SrO
3. Beryllium nitride — Be_3N_2
4. Calcium fluoride — CaF_2
5. Potassium chloride — KCl
6. Cesium Phosphide — Cs_3P
7. Barium Bromide — $BaBr_2$
8. Lithium Oxide — Li_2O
9. Aluminum Selenide — Al_2Se_3
10. Rubidium Iodide — RbI

What is a Polyatomic Ion?

A polyatomic ion is an ion made from a charged group of bonded atoms consisting of more than one element.

Watch the following instructional videos:

Definition: https://www.youtube.com/watch?v=MJZeZvDxcx8

IMPORTANT: In AP Chemistry, **the names, formulas, and charges of the common polyatomic ions MUST BE MEMORIZED! You will be tested on these during the first two days of school.**

AP Chemistry Polyatomic Ion Memorization List

(see the next page for tips on how to do it!)

1+ Cations

NH_4^{1+}	Ammonium

1- Anions

$C_2H_3O_2^{1-}$	Acetate
CN^{1-}	Cyanide
BrO^{1-}	Hypobromite
BrO_2^{1-}	Bromite
BrO_3^{1-}	Bromate
BrO_4^{1-}	Perbromate
ClO^{1-}	Hypochlorite
ClO_2^{1-}	Chlorite
ClO_3^{1-}	Chlorate
ClO_4^{1-}	Perchlorate
IO^{1-}	Hypoiodite
IO_2^{1-}	Iodite
IO_3^{1-}	Iodate
IO_4^{1-}	Periodate
HCO_3^{1-}	Hydrogen Carbonate
$H_2PO_4^{1-}$	Dihydrogen Phosphate
HSO_4^{1-}	Hydrogen Sulfate
NO_2^{1-}	Nitrite
NO_3^{1-}	Nitrate
OH^{1-}	Hydroxide
MnO_4^{1-}	Permanganate
SCN^{1-}	thiocyanate

2- Anions

CO_3^{2-}	Carbonate
$C_2O_4^{2-}$	Oxalate
CrO_4^{2-}	Chromate
$Cr_2O_7^{2-}$	Dichromate
HPO_4^{2-}	Hydrogen Phosphate
O_2^{2-}	Peroxide
SO_3^{2-}	Sulfite
SO_4^{2-}	Sulfate
$S_2O_3^{2-}$	Thiosulfate

3- Anions

PO_3^{3-}	Phosphite
PO_4^{3-}	Phosphate

A Strategy for Memorizing the Polyatomic Ions

Watch this video explanation:

https://www.youtube.com/watch?v=cpaQhgNgF7U&feature=youtu.be

The Big Five (make flashcards for these!)

1. CO_3^{2-} carbonate
2. ClO_3^{1-} chlorate
3. NO_3^{1-} nitrate
4. SO_4^{2-} sulfate
5. PO_4^{3-} phosphate

Memorization Tip:

Slivka's Square (with corners: Si, S, Te, Sn)

- inside square: -ate has 4 O
- outside square: -ate has 3 O

Ions by Analogy

If you know…

Cl ClO_3^{1-}

you know…

Br and I BrO_3^{1-} and IO_3^{1-}

Ions by Extension

Memorize the –ate ion (example: Chlorate, ClO_3) and you can figure out the rest…

- Hypo- ___ -ite is under –ite (ClO^{1-}) think: hypodermic needle
- ___ -ite is one less (ClO_2^{1-}) - goes under the skin
- ___ -ate memorize this (ClO_3^{1-})
- Per- ___ -ate is over –ate (ClO_4^{1-}) think: hyper – overexcited

Ions by Hydrogenation

SO_4^{2-}	sulfate	CO_3^{2-}	carbonate
HSO_4^{1-}	hydrogen sulfate	HCO_3^{1-}	hydrogen carbonate
PO_4^{3-}	phosphate		
HPO_4^{2-}	hydrogen phosphate		
$H_2PO_4^{1-}$	dihydrogen phosphate		

Little Five (make flashcards for these!)

1. Polyatomic Cations
   NH_4^{1+} ammonium
2. Named like a Monotomic Anion (-ide ending)
   OH^{1-} hydroxide
   CN^{1-} cyanide
   O_2^{2-} peroxide
3. Colored Anions
   MnO_4^{1-} permanganate (purple)
   CrO_4^{2-} chromate (yellow)
   $Cr_2O_7^{2-}$ dichromate (orange)
4. Organic Polyatomic Anions
   $C_2H_3O_2^{1-}$ acetate
   $C_2O_4^{2-}$ oxalate
5. Thio- Anions
   SCN^{1-} thiocyanate
   $S_2O_3^{2-}$ thiosulfate

Naming & Writing Formulas of Ionic Compounds that contain Polyatomic Ions

Naming Ionic Compounds that contain Polyatomic Ions

General Rule:

1. Identify that the ionic compound contains a polyatomic ion. Ionic compounds containing polyatomic ions are easily identified because they will contain more than two elements.
2. Identify the particular polyatomic ion the compound contains. Unless it is ammonia (NH_4^+), the polyatomic ion will be negatively charged, and therefore, will be written AFTER the metal ion (which, by convention, is always written first).
3. Simply write the name of the metal ion (same as the element's name) and then the name of the polyatomic ion (which you've memorized).

Example: $MgCO_3$ Mg^{2+} and CO_3^{2-} → Magnesium Carbonate

Writing Formulas for Ionic Compounds that contain Polyatomic Ions

Watch the following instructional videos:

Definition: https://www.youtube.com/watch?v=MJZeZvDxcx8

Introduction: https://www.youtube.com/watch?v=p9iQ5Qn42DM

Practice Problems

Instructions: Write the formula from the names of the following ionic compounds and vice versa.

1. _____ Na_2SO_3
2. _____ $Sr(NO_3)_2$
3. _____ $Be_3(PO_4)_2$
4. _____ NH_4F
5. Cesium Acetate _____
6. Barium Perchlorate _____
7. Ammonium Cyanide _____
8. Aluminum Hydroxide _____

Naming & Writing Formulas of Ionic Compounds that contain Polyatomic Ions

ANSWERS

Practice Problems

Instructions: Check your answers

1. Sodium Sulfite — Na_2SO_3
2. Strontium Nitrate — $Sr(NO_3)_2$
3. Beryllium Phosphate — $Be_3(PO_4)_2$
4. Ammonium Fluoride — NH_4F
5. Cesium Acetate — $CsC_2H_3O_2$
6. Barium Perchlorate — $Ba(ClO_4)_2$
7. Ammonium Cyanide — NH_4CN
8. Aluminum Hydroxide — $Al(OH)_3$

Naming & Writing Formulas of Ionic Compounds that contain Transition Metals

Writing Formulas of Ionic Compounds that contain Transition Metals

Watch the following instructional videos:

Definition: https://www.youtube.com/watch?v=Da_ah6TqAss

Introduction: https://www.youtube.com/watch?v=H4nfE8gRX7Q

Naming Ionic Compounds that contain Transition Metals

Watch the following instructional videos:

Introduction: https://www.youtube.com/watch?v=Rq0A-AHdB74

Practice: https://www.youtube.com/watch?v=Rq0A-AHdB74

Instructions: Write the formula from the names of the following binary ionic compounds and vice versa.

1. _____ Mn_2S
2. _____ NiO
3. _____ Cr_3N_2
4. _____ CuF_2
5. _____ PbCl
6. Tin II Phosphide _____
7. Molybdenum III Bromide _____
8. Titanium IV Oxide _____
9. Vanadium II Selenide _____
10. Cadmium I Sulfate _____

Naming & Writing Formulas of Ionic Compounds that contain Transition Metals

ANSWERS

Instructions: Check your answers.

1. Manganese I Sulfide — Mn_2S
2. Nickel II Oxide — NiO
3. Chromium II Nitride — Cr_3N_2
4. Copper II Fluoride — CuF_2
5. Lead I Chloride — PbCl
6. Tin II Phosphide — Sn_3P_2
7. Molybdenum III Bromide — $MoBr_3$
8. Titanium IV Oxide — TiO_2
9. Vanadium II Selenide — VSe
10. Cadmium I Sulfate — Cd_2SO_4

Writing Names & Formulas of Acids

Identifying an Acid

An acid **starts with hydrogen** and ends with a nonmetal or polyatomic ion.

Watch the following instructional videos:

Introduction: https://www.youtube.com/watch?v=5Jb2u9ihfm4

Practice: https://www.youtube.com/watch?v=VyjnMk-Ta10

General Rules:

Binary acids: acids made from hydrogen plus a single element (like HBr):

Hydro + element name + ic Acid

Example: HBr Hydro + brom~~ine~~ Acid → Hydrobromic Acid

Oxyacids: acids made from hydrogen plus an oxygen-containing polyatomic ion (like $HBrO_3$):

Polyatomic ion name* Acid
(*but change "ate" endings to "ic" and "ite" endings to "ous")

Example:	$HBrO_3$	BrO_3^- = Bromate	Brom~~ate~~ Acid	→	Bromic Acid
	$HBrO_2$	BrO_2^- = Bromite	Brom~~ite~~ Acid	→	Bromous Acid

Practice Problems

Instructions: Write the formulas from the names of the following acids and vice versa.

1. _____ HCl
2. _____ HI
3. _____ H_2SO_3
4. _____ HNO_3
5. Phosphorous Acid _____
6. Hydrofluoric Acid _____
7. Perchloric Acid _____
8. Acetic Acid _____

Writing Names & Formulas of Acids

ANSWERS

Practice Problems

Instructions: Check your answers.

1. Hydrochloric Acid — HCl
2. Hydroiodic Acid — HI
3. Sulfurous Acid — H_2SO_3
4. Nitric Acid — HNO_3
5. Phosphoric Acid — H_3PO_4
6. Hydrofluoric Acid — HF
7. Hypochlorous Acid — HClO
8. Acetic Acid — $HC_2H_3O_2$

Naming Hydrates

Hydrates are ionic compounds that have a specific number of water molecules attached to them. **Anhydrates** are hydrates that have had the water removed (such as through heating). The number of water is indicated with a coefficient in the formula and a prefix in the name.

Examples of common hydrates

Formula	Name
$CuSO_4 \cdot 5H_20$	copper II sulfate pentahydrate
$LiCl \cdot H_2O$	lithium chloride monohydrate
$BaCl_2 \cdot 2H_2O$	barium chloride dehydrate
$MgSO_4 \cdot 7H_2O$	magnesium sulfate heptahydrate
$Sr(NO_3)_2 \cdot 4H_2O$	strontium nitrate tetrahydrate

Practice Problems

Instructions: Write the formulas from the names of the following hydrates and vice versa.

1. _____ $MgBr_2 \cdot 2H_2O$
2. _____ $CoCl_2 \cdot 6H_2O$
3. _____ $NaNO_3 \cdot 5H_2O$
4. _____ $NH_4ClO_2 \cdot 3H_2O$
5. potassium sulfide tetrahydrate _____
6. lithium hydroxide monohydrate _____
7. calcium sulfide octahydrate _____
8. barium chloride heptahydrate _____

Naming Hydrates
ANSWERS

Practice Problems

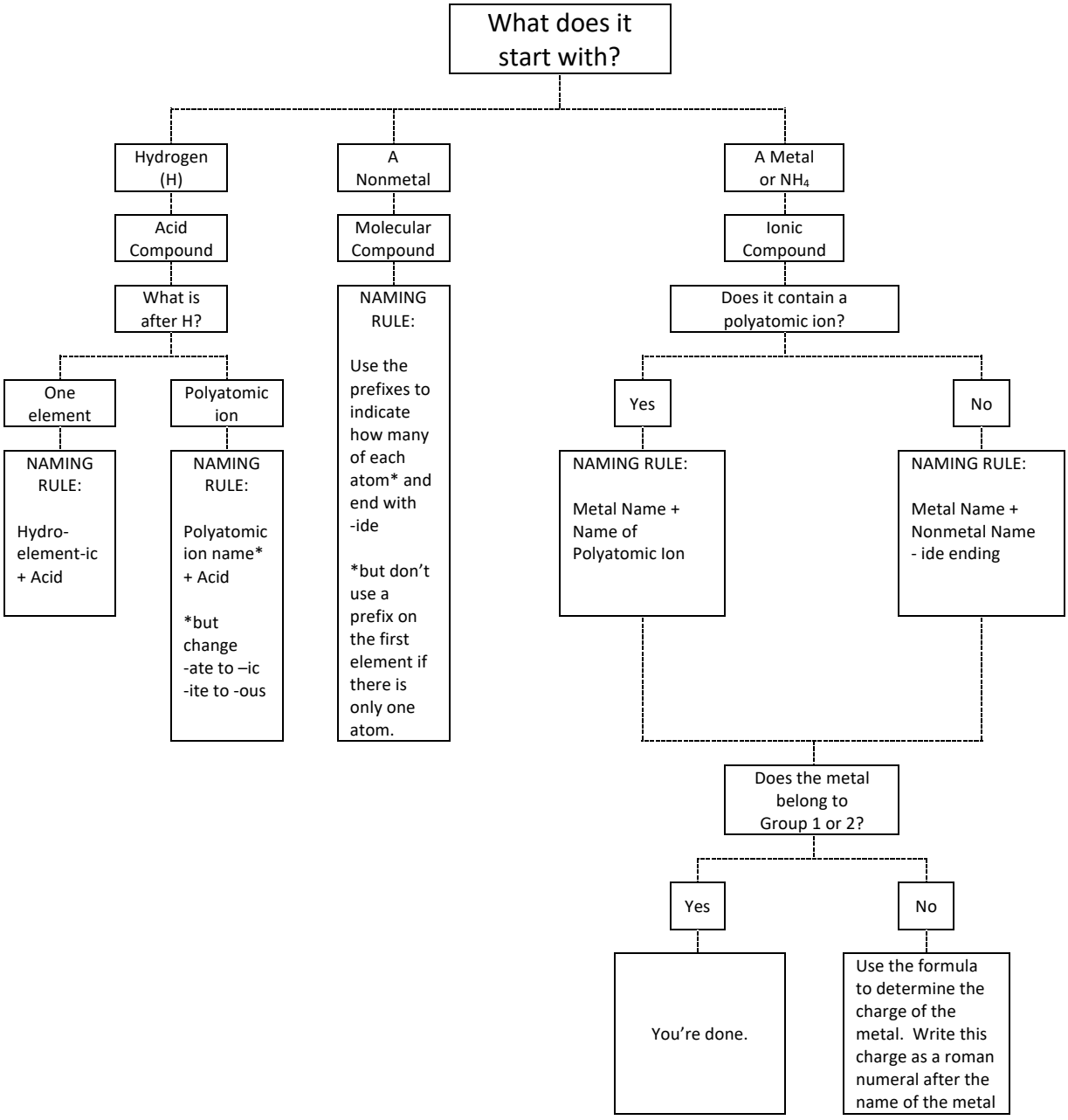
Instructions: Write the formulas from the names of the following hydrates and vice versa.

1. magnesium bromide dihydrate — $MgBr_2 \cdot 2H_2O$
2. cobalt II chloride hexahydrate — $CoCl_2 \cdot 6H_2O$
3. sodium nitrate pentahydrate — $NaNO_3 \cdot 5H_2O$
4. ammonium chlorite trihydrate — $NH_4ClO_2 \cdot 3H_2O$
5. potassium sulfide tetrahydrate — $K_2S \cdot 4H_2O$
6. lithium sulfate monohydrate — $Li_2SO_4 \cdot H_2O$
7. calcium carbonate octahydrate — $CaCO_3 \cdot 8H_2O$
8. barium chloride heptahydrate — $BaCl_2 \cdot 7H_2O$

Putting It All Together

Of course, in chemistry class, you will be asked to name and write the formulas of random compounds. They won't come pre-categorized like the ones on the previous pages. So let's come up with a strategy for how to correctly name or write the formula of a compound when you haven't been told what kind it is.

When you come across a random formula, answer the following questions:



Summary of Naming and Writing Formula Rules

Type of Compound	Naming the Compound	Writing the Formula
Ionic ***(metal + nonmetal)***	**Metal Name[1] + Nonmetal[2] + *ide*** 1) Most metals that are not in Group 1 & 2 can have more than one charge. For these, determine the charge of the ion by determining what the charge on each metal would have to be to make the compound neutral, by considering the charge of the negative ion and the entire formula. Indicate the metals charge with a roman numeral (I, II, III, IV). 2) If the compound in not binary (has more than two element), find the polyatomic ion and name it.	***Using the charge of each ion in the formula, determine the number of each ion (ratio) needed in order to create a neutral compound.***
Molecular ***(all nonmetals)***	***Prefix* + Nonmetal +** ***Prefix* + Nonmetal + *ide*** Remember: no *mono* is needed for the first element.	***Write the formula using the prefixes provided in the name to determine how many atoms of each element there should be.***
Acid ***(Begins with H)***	*Binary Acid (Without Oxygen)* **Hydro + nonmetal + ic + Acid** *Oxyacid (With Oxygen)* **Polyatomic Ion Name[3] + Acid** 3) replace an **–ite** ending with **–ous** **-ate** ending with **–ic**	***Using the charge of each ion in the formula, determine the number of each ion (ratio) needed in order to create a neutral compound.***

Practice Writing Formulas

Mix of Molecular, Ionic, and Acid Compounds

(tip: use the flow chart strategy above to solve these!)

1) aluminum arsenide _____
2) permanganic acid _____
3) tin II sulfite _____
4) vanadium (V) fluoride _____
5) lithium acetate _____
6) hexaboron monosilicide _____
7) hydrofluoric acid _____
8) ammonium nitrate _____
9) silver cyanide _____
10) copper (II) chloride _____
11) phosphorus triiodide _____
12) potassium fluoride _____
13) hypobromous acid _____
14) nitric acid _____
15) acetic acid _____
16) phosphoric acid _____
17) chlorine dioxide _____
18) periodic acid _____
19) potassium permanganate _____
20) chromium (VI) cyanide _____
21) sulfuric acid _____
22) perchloric acid _____
23) vanadium (III) selenide _____

24) manganese (IV) nitride _____

25) beryllium oxide _____

26) sodium sulfate _____

27) iodous acid _____

28) sulfurous acid _____

29) copper (II) bicarbonate _____

30) iodine pentafluoride _____

31) dinitrogen trioxide _____

32) bromous acid _____

33) chlorous acid _____

34) ammonium sulfate _____

35) magnesium iodide _____

36) copper (II) sulfite _____

37) aluminum phosphate _____

38) lead (II) nitrite _____

39) cobalt (II) selenide _____

40) iron (II) oxide _____

41) lithium cyanide _____

42) lead (IV) sulfite _____

43) phosphorous acid _____

Practice Naming Compounds

Mix of Molecular, Ionic, and Acid Compounds

1) P_4S_5 _____

2) HI _____

3) H_2SO_4 _____

4) SeF_6 _____

5) SrS _____

6) Cu_2S _____

7) Si_2Br_6 _____

8) $HBrO_4$ _____

9) $HBrO_3$ _____

10) $HC_2H_3O_2$ _____

11) SCl_4 _____

12) ZnI_2 _____

13) NaF _____

14) K_2CO_3 _____

15) $Ca_3(PO_4)_2$ _____

16) NH_4I _____

17) H_3PO_3 _____

18) H_3P _____

19) CH_4 _____

20) B_2Si _____

21) Cu_2S _____

22) H_2CrO_4 _____

23) SrS _____

24) $MgCl_2$ _____

25) HIO_3 _____

26) H_2S _____

27) $Be(OH)_2$ _____

28) $Mn(NO_3)_3$ _____

29) HCN _____

30) H_2CO_3 _____

31) $FePO_4$ _____

32) $CoCO_3$ _____

33) HCl _____

34) $HClO_4$ _____

35) NF_3 _____

36) $AgNO_3$ _____

Practice Writing Formulas ANSWERS

Mix of Molecular, Ionic, and Acid Compounds

1) aluminum arsenide — $AlAs$
2) permanganic acid — $HMnO_4$
3) tin (II) sulfite — $SnSO_3$
4) vanadium (V) fluoride — VF_5
5) lithium acetate — $LiC_2H_3O_2$
6) hexaboron monosilicide — B_6Si
7) hydrofluoric acid — HF
8) ammonium nitrate — NH_4NO_3
9) silver (I) cyanide — $AgCN$
10) copper (II) chloride — $CuCl_2$
11) phosphorus triiodide — PI_3
12) potassium fluoride — KF
13) hypobromous acid — $HBrO$
14) nitric acid — HNO_3
15) acetic acid — $HC_2H_3O_2$
16) phosphoric acid — H_3PO_4
17) chlorine dioxide — ClO_2
18) periodic acid — HIO_4
19) potassium permanganate — $KMnO_4$
20) chromium (VI) cyanide — $Cr(CN)_6$
21) sulfuric acid — H_2SO_4
22) perchloric acid — $HClO_4$
23) vanadium (III) selenide — V_2Se_3

24)	manganese (IV) nitride	Mn_3N_4
25)	beryllium oxide	BeO
26)	sodium sulfate	Na_2SO_4
27)	iodous acid	HIO_2
28)	sulfurous acid	H_2SO_3
29)	copper (II) hydrogen carbonate	$Cu(HCO_3)_2$
30)	iodine pentafluoride	IF_5
31)	dinitrogen trioxide	N_2O_3
32)	bromous acid	$HBrO_2$
33)	chlorous acid	$HClO_2$
34)	ammonium sulfate	$(NH_4)_2SO_4$
35)	magnesium iodide	MgI_2
36)	copper (II) sulfite	$CuSO_3$
37)	aluminum phosphate	$AlPO_4$
38)	lead (II) nitrite	$Pb(NO_2)_2$
39)	cobalt (II) selenide	$CoSe$
40)	iron (II) oxide	FeO
41)	lithium cyanide	$LiCN$
42)	lead (IV) sulfite	$Pb(SO_3)_2$
43)	phosphorous acid	H_3PO_3

Practice Naming Compounds ANSWERS

Mix of Molecular, Ionic, and Acid Compounds

1) P_4S_5 tetraphosphorous pentasulfide
2) HI hydroiodic acid
3) H_2SO_4 sulfuric acid
4) SeF_6 selenium hexafluoride
5) SrS strontium sulfide
6) Cu_2S copper (I) sulfide
7) Si_2Br_6 disilicon hexabromide
8) $HBrO_4$ perbromic acid
9) $HBrO_3$ bromic acid
10) $HC_2H_3O_2$ acetic acid
11) SCl_4 sulfur tetrachloride
12) ZnI_2 zinc (II) iodide
13) NaF sodium flouride
14) K_2CO_3 potassium carbonate
15) $Ca_3(PO_4)_2$ calicum phosphate
16) NH_4I ammonium iodide
17) H_3PO_3 phosphorous acid
18) H_3P hydrophosphoric acid
19) CH_4 carbon tetrahydride (or methane)
20) B_2Si diboron monosilicide
21) Cu_2S copper (I) sulfide
22) H_2CrO_4 chromic acid
23) SrS strontium sulfide

24)	$MgCl_2$	magnesium chloride
25)	HIO_3	iodic acid
26)	H_2S	hydrosulfuric acid
27)	$Be(OH)_2$	beryllium hydroxide
28)	$Mn(NO_3)_3$	manganese (III) nitrate
29)	HCN	cyanic acid
30)	H_2CO_3	carbonic acid
31)	$FePO_4$	iron (III) phosphate
32)	$CoCO_3$	cobalt (II) carbonate
33)	HCl	hydrochloric acid
34)	$HClO_4$	perchloric acid
35)	NF_3	nitrogen trifluoride
36)	$AgNO_3$	silver (I) nitrite